



Wazuh SIEM–Practical

Security Monitoring Project

Log Collection | FIM | Custom Rules | Alerting

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Project Type: Hands-On-SIEM Lab

This project demonstrates the practical implementation of Wazuh SIEM for monitoring system activity, detecting security events, and generating alerts

1. Project Overview

The purpose of this project was to build a basic SOC-style monitoring environment using wazuh SIEM.

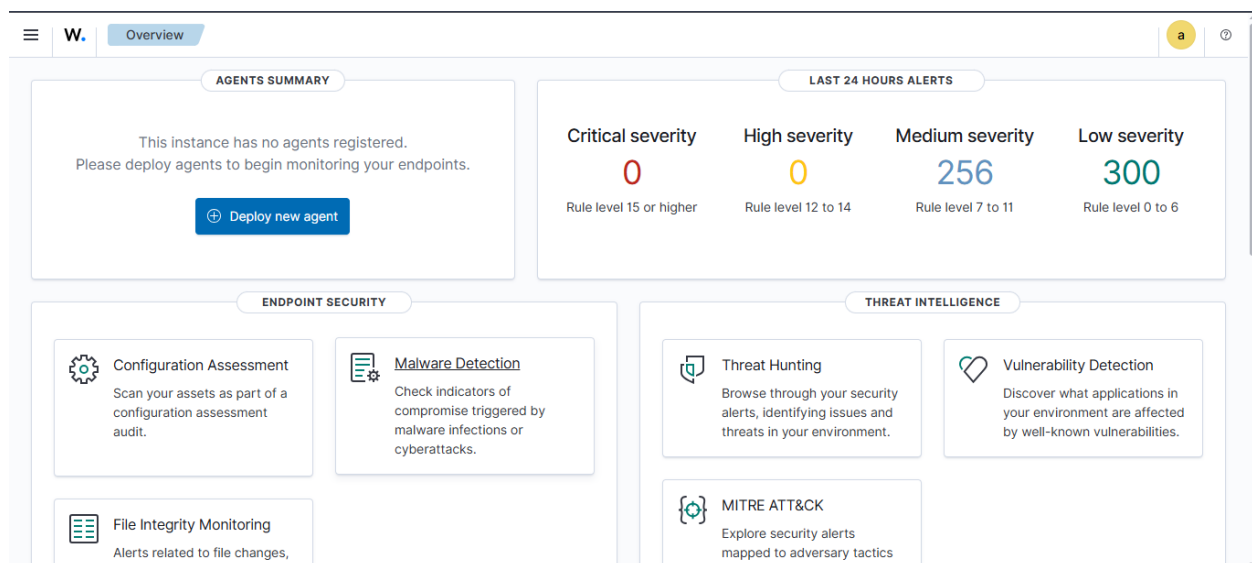
The focus was to understand how security events are collected from an endpoint, detected using rules, and then converted into alerts for investigation.

This project covers:

- Endpoint onboarding using wazuh agent
- File Integrity Monitoring (FIM)
- Creating Custom Rules
- Email alert notification

Tools & Environment

- Wazuh Manager (Ubuntu 20.04)
- Windows 10 endpoint
- Gmail SMTP (for alerts)



2. Agent Deployment (Windows Endpoint)

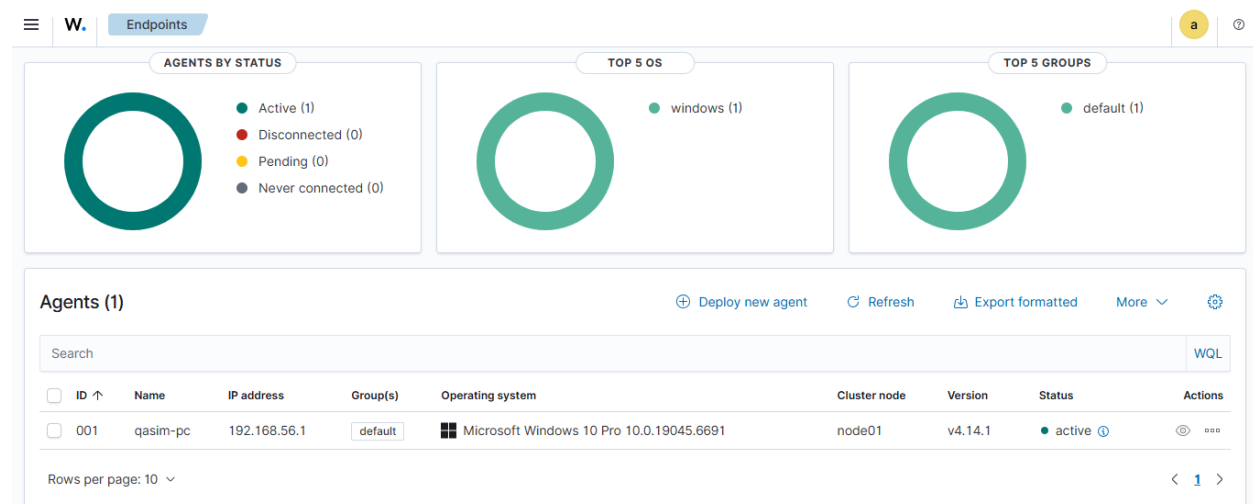
To start monitoring a wazuh agent was installed on a Windows 10 system.

The agent was connected to the wazuh manager and verified as active from the dashboard.

Commands :

1. `msiexec.exe /i wazuh-agent-4.14.1-1.msi /q`
`WAZUH_MANAGER="192.168.56.106"`
`WAZUH_AGENT_NAME="qasim-pc"`
2. `NET START Wazuh`

After installation, the agent status showed Active, confirming successful communication.



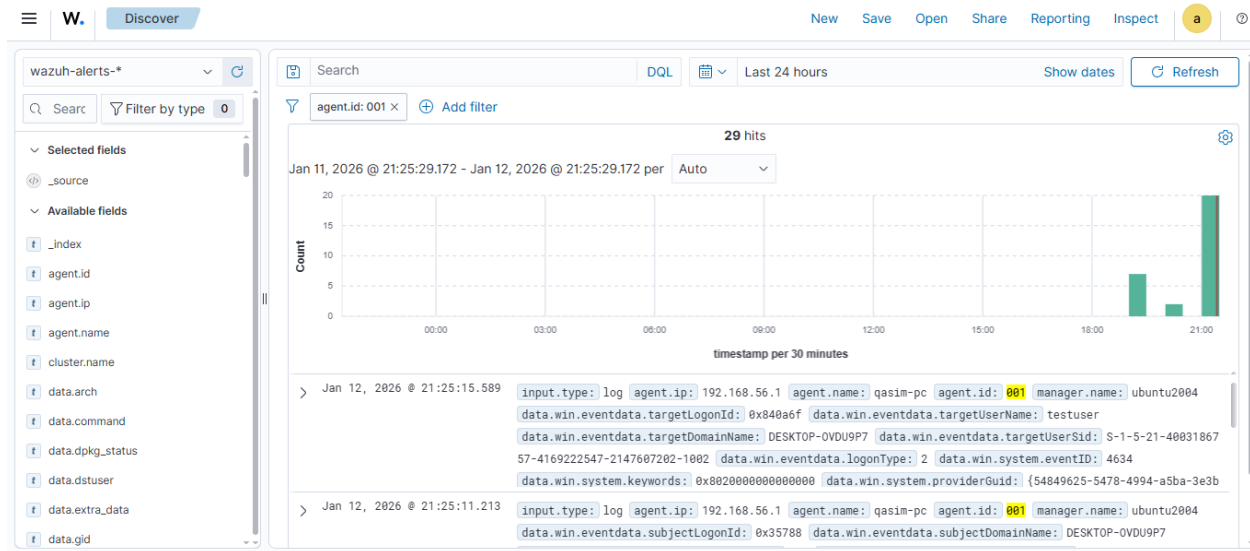
3. Log Collection Verification

After installing the agent, I verified that logs from the Windows system were successfully reaching the wazuh SIEM

In the Discover section of the dashboard, Windows security logs such as:

- Login attempts
- User activity

- System events were clearly visible and searchable.



Logs are successfully received and searchable in Wazuh

4. File Integrity Monitoring (FIM)

File integrity monitoring (FIM) was configured to monitor a sensitive directory on the Windows endpoint.

A folder named C:\Secure_assets was defined as a monitored location in wazuh. The purpose was to detect any unauthorized file activity inside this directory.

FIM Configuration (ossec.conf):

```

<!-- File integrity monitoring -->
<syscheck>

  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>300</frequency>

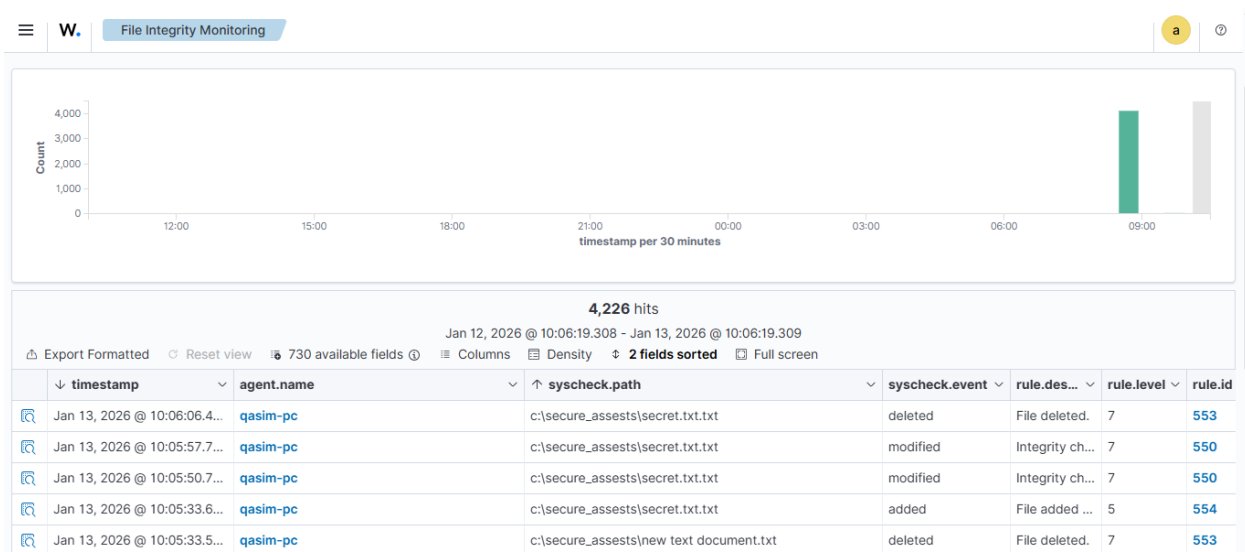
  <!-- Default files to be monitored. -->
  <directories>C:\Secure_assets</directories>

```

To validate FIM, different file activities were performed inside the monitored folder:

- A file was created
- The file content was modified
- The file was deleted

For each of these actions wazuh generated alerts showing the file path, type of change, timestamp, and the endpoint where the activity occurred.

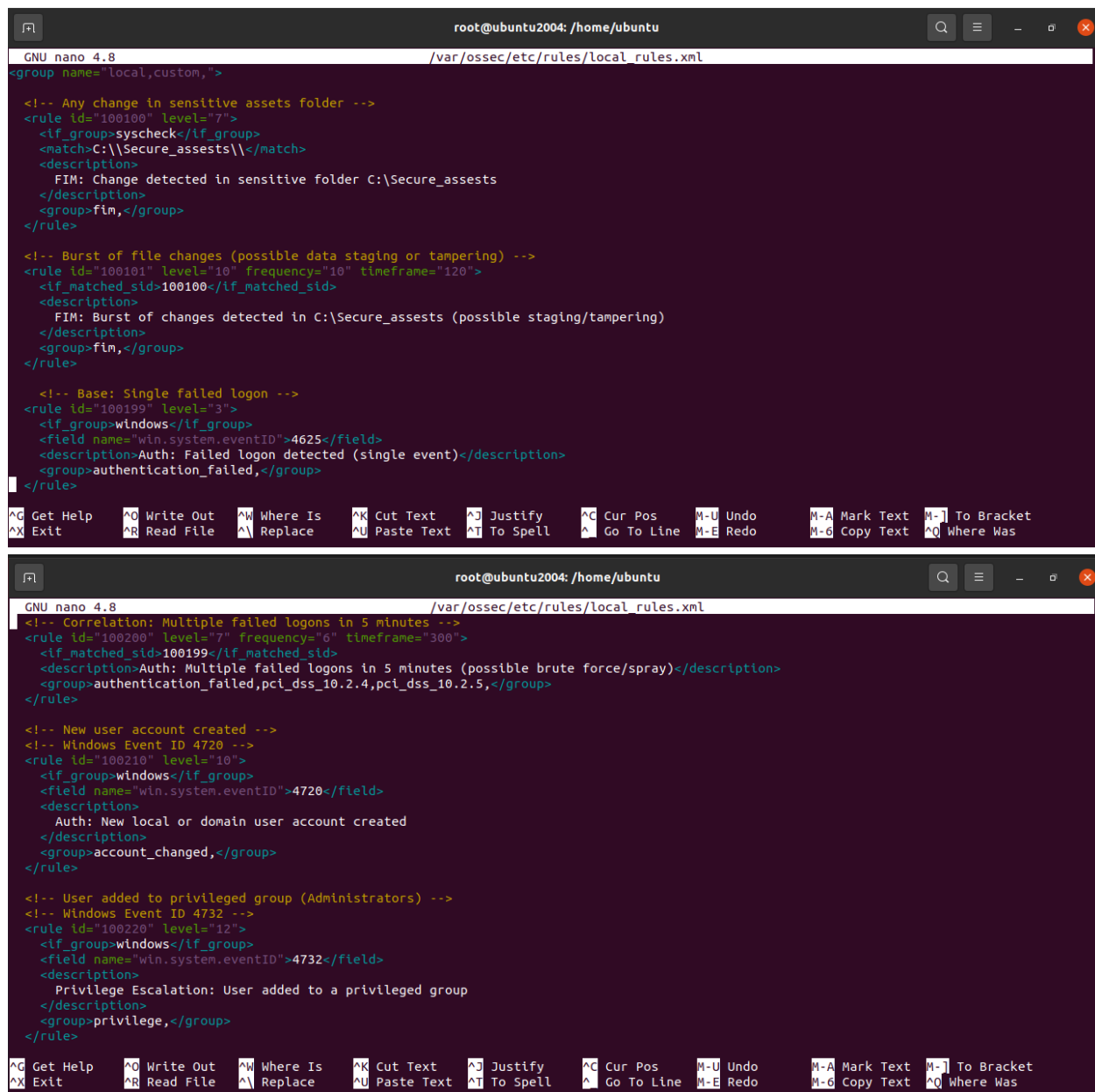


This confirmed that File Integrity Monitoring was properly configured and working as expected for sensitive data monitoring

5. Custom Detection Rules

In addition to the default wazuh rules custom detection rules were created to support basic SOC use cases. These rules focused on detecting sensitive folder changes, authentication misuse, privilege escalation, and persistence-related activity.

The rules were written in the local_rules.xml file and tested using real actions on the Windows endpoint



```
root@ubuntu2004: /home/ubuntu
GNU nano 4.8 /var/ossec/etc/rules/local_rules.xml
<group name="local,custom,">

  <!-- Any change in sensitive assets folder -->
  <rule id="100100" level="7">
    <if_group>syscheck</if_group>
    <match>C:\\Secure_assests\\</match>
    <description>
      FIM: Change detected in sensitive folder C:\\Secure_assests
    </description>
    <group>fim,</group>
  </rule>

  <!-- Burst of file changes (possible data staging or tampering) -->
  <rule id="100101" level="10" frequency="10" timeframe="120">
    <if_matched_sid>100100</if_matched_sid>
    <description>
      FIM: Burst of changes detected in C:\\Secure_assests (possible staging/tampering)
    </description>
    <group>fim,</group>
  </rule>

  <!-- Base: Single failed logon -->
  <rule id="100199" level="3">
    <if_group>windows</if_group>
    <field name="win.system.eventID">4625</field>
    <description>Auth: Failed logon detected (single event)</description>
    <group>authentication_failed,</group>
  </rule>

  <!-- Correlation: Multiple failed logons in 5 minutes -->
  <rule id="100200" level="7" frequency="6" timeframe="300">
    <if_matched_sid>100199</if_matched_sid>
    <description>Auth: Multiple failed logons in 5 minutes (possible brute force/spray)</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>

  <!-- New user account created -->
  <!-- Windows Event ID 4720 -->
  <rule id="100210" level="10">
    <if_group>windows</if_group>
    <field name="win.system.eventID">4720</field>
    <description>
      Auth: New local or domain user account created
    </description>
    <group>account_changed,</group>
  </rule>

  <!-- User added to privileged group (Administrators) -->
  <!-- Windows Event ID 4732 -->
  <rule id="100220" level="12">
    <if_group>windows</if_group>
    <field name="win.system.eventID">4732</field>
    <description>
      Privilege Escalation: User added to a privileged group
    </description>
    <group>privilege,</group>
  </rule>

</group>
```

```
root@ubuntu2004: /home/ubuntu
GNU nano 4.8 /var/ossec/etc/rules/local_rules.xml
Privilege Escalation: User added to a privileged group
</description>
<group>privilege,</group>
</rule>

<!-- Scheduled task created (common persistence technique) -->
<!-- Windows Event ID 4698 -->
<rule id="100230" level="10">
  <if_group>windows</if_group>
  <field name="win.system.eventID">4698</field>
  <description>
    Persistence: New scheduled task created
  </description>
  <group>persistence,</group>
</rule>

<!-- Suspicious PowerShell execution flags -->
<rule id="100240" level="9">
  <if_group>sysmon</if_group>
  <match>-enc|-EncodedCommand|-w hidden|-WindowStyle Hidden</match>
  <description>
    Execution: Suspicious PowerShell flags detected (encoded or hidden)
  </description>
  <group>execution,powershell,</group>
</rule>
</group>

^G Get Help      ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify      ^C Cur Pos      M-U Undo        M-A Mark Text   M-] To Bracket
^X Exit          ^R Read File    ^\ Replace      ^U Paste Text   ^T To Spell     ^_ Go To Line    M-E Redo        M-G Copy Text   ^Q Where Was
```

These helped in understanding how custom SOC detections are created and validated

6. Attack Simulation & Alert Generation

After creating the custom rules, different test activities were performed on the Windows endpoint to simulate attacker behavior and validate detections.

The following actions were used to generate alerts in wazuh:

- Creating, modifying, and deleting files inside the monitored sensitive folder
- Performing multiple failed login attempts to simulate authentication abuse
- Creating a new user and adding it to the Administrators group to simulate privilege escalation
- Creating a scheduled task to simulate persistence

For each activity, wazuh successfully generated alerts with appropriate severity levels, confirming that the custom rules were working as expected

```
PS C:\Windows\system32> echo "secret data" > C:\Secure_assests\file1.txt
PS C:\Windows\system32> echo "secret data" > C:\Secure_assests\file1.txt
PS C:\Windows\system32> echo "updated" >> C:\Secure_assests\file1.txt
PS C:\Windows\system32> Remove-Item C:\Secure_assests\file1.txt
PS C:\Windows\system32> 1..12 | % { "data $_" | Out-File "C:\Secure_assests\doc_$_.txt" }
PS C:\Windows\system32> net user ghost Ghost@123 /add
The command completed successfully.

PS C:\Windows\system32> net localgroup administrators ghost /add
The command completed successfully.

PS C:\Windows\system32> powershell -EncodedCommand RwB1AHQALQBQAHIAbwBjAGUAcwBzAA==
```

```
PS C:\Users\Jannat Enterprises> runas /user:ghost powershell
Enter the password for ghost:
Attempting to start powershell as user "DESKTOP-OVDU9P7\ghost" ...
RUNAS ERROR: Unable to run - powershell
1326: The user name or password is incorrect.

PS C:\Users\Jannat Enterprises> runas /user:ghost powershell
Enter the password for ghost:
Attempting to start powershell as user "DESKTOP-OVDU9P7\ghost" ...
RUNAS ERROR: Unable to run - powershell
1326: The user name or password is incorrect.

PS C:\Users\Jannat Enterprises> runas /user:ghost powershell
Enter the password for ghost:
Attempting to start powershell as user "DESKTOP-OVDU9P7\ghost" ...
RUNAS ERROR: Unable to run - powershell
1326: The user name or password is incorrect.

PS C:\Users\Jannat Enterprises> runas /user:ghost powershell
Enter the password for ghost:
Attempting to start powershell as user "DESKTOP-OVDU9P7\ghost" ...
RUNAS ERROR: Unable to run - powershell
1326: The user name or password is incorrect.

PS C:\Users\Jannat Enterprises> runas /user:ghost powershell
Enter the password for ghost:
Attempting to start powershell as user "DESKTOP-OVDU9P7\ghost" ...
RUNAS ERROR: Unable to run - powershell
1326: The user name or password is incorrect.
```

Generated Alerts:

≡	W.	Threat Hunting	a	🔍
↓ timestamp	agent.name	rule.description	rule.level	rule.id
Jan 13, 2026 @ 11:02:00.1...	qasim-pc	Wazuh agent stopped.	3	506
Jan 13, 2026 @ 10:59:54.2...	qasim-pc	Non service account logged off.	3	67023
Jan 13, 2026 @ 10:59:54.2...	qasim-pc	Non service account logged off.	3	67023
Jan 13, 2026 @ 10:59:02.9...	qasim-pc	Special privileges assigned to new logon.	3	67028
Jan 13, 2026 @ 10:59:02.9...	qasim-pc	Windows Workstation Logon Success	3	60118
Jan 13, 2026 @ 10:59:02.9...	qasim-pc	Windows Workstation Logon Success	3	60118
Jan 13, 2026 @ 10:51:15.4...	qasim-pc	Special privileges assigned to new logon.	3	67028
Jan 13, 2026 @ 10:51:15.4...	qasim-pc	Windows Workstation Logon Success	3	60118
Jan 13, 2026 @ 10:51:15.4...	qasim-pc	Windows Workstation Logon Success	3	60118
Jan 13, 2026 @ 10:51:05.0...	qasim-pc	Auth: Failed logon detected (single event)	3	100199
Jan 13, 2026 @ 10:51:00.1...	qasim-pc	Auth: Failed logon detected (single event)	3	100199
Jan 13, 2026 @ 10:50:57.2...	qasim-pc	Auth: Multiple failed logons in 5 minutes (possible brute force/spray)	7	100200
Jan 13, 2026 @ 10:50:54.7...	qasim-pc	Auth: Failed logon detected (single event)	3	100199
Jan 13, 2026 @ 10:50:52.1...	qasim-pc	Auth: Failed logon detected (single event)	3	100199
Jan 13, 2026 @ 10:50:49.7...	qasim-pc	Auth: Failed logon detected (single event)	3	100199

≡	W.	Threat Hunting	a	🔍
↓ timestamp	agent.name	rule.description	rule.level	rule.id
Jan 13, 2026 @ 10:50:46.8...	qasim-pc	Auth: Failed logon detected (single event)	3	100199
Jan 13, 2026 @ 10:50:39.8...	qasim-pc	Auth: Failed logon detected (single event)	3	100199
Jan 13, 2026 @ 10:47:57.6...	qasim-pc	Privilege Escalation: User added to a privileged group	12	100220
Jan 13, 2026 @ 10:46:38.2...	qasim-pc	Privilege Escalation: User added to a privileged group	12	100220
Jan 13, 2026 @ 10:46:38.2...	qasim-pc	User account changed	8	60110
Jan 13, 2026 @ 10:46:38.2...	qasim-pc	User account enabled or created	8	60109
Jan 13, 2026 @ 10:46:38.2...	qasim-pc	Auth: New local or domain user account created	10	100210
Jan 13, 2026 @ 10:46:38.1...	qasim-pc	Domain Users Group Changed	5	60160
Jan 13, 2026 @ 10:44:04.4...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:44:04.4...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:44:04.4...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:44:04.3...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:44:04.3...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:44:04.3...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:44:04.2...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100

≡	W.	Threat Hunting	a	🔍
↓ timestamp	agent.name	rule.description	rule.level	rule.id
Jan 13, 2026 @ 10:44:04.2...	qasim-pc	FIM: Burst of changes detected in C:\Secure_assests (possible staging/tampering)	10	100101
Jan 13, 2026 @ 10:44:04.2...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:44:04.1...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:44:04.1...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:44:04.1...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:43:54.4...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:43:44.9...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:43:31.9...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:43:31.5...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100
Jan 13, 2026 @ 10:43:30.5...	qasim-pc	FIM: Change detected in sensitive folder C:\Secure_assests	7	100100

7.Email Alerting (SOC Notification)

In a real SOC environment, critical security alerts are sent through email so analysts can respond quickly. For this project, email alerting was configured using Gmail SMTP to simulate this process.

To generate alerts, a brute-force login attempt was performed on the Windows endpoint. Once the alert severity reached the defined threshold, wazuh automatically sent an email notification.

Only alerts with a severity level of 8 or higher were configured to trigger email notifications, ensuring that only important security events were sent.

Email Configuration:

```
<ossec_config>
<global>
  <jsonout_output>yes</jsonout_output>
  <alerts_log>yes</alerts_log>
  <logall>no</logall>
  <logall_json>no</logall_json>
  <email_notification>yes</email_notification>
  <email_notification>127.0.0.1</smtp_server>
  <email_from>wazuh-alerts@wazuh.com</email_from>
  <email_to>qasim.wazuh12@gmail.com</email_to>
  <email_maxperhour>12</email_maxperhour>
  <email_log_source>alerts.log</email_log_source>
  <agents_disconnection_time>15m</agents_disconnection_time>
  <agents_disconnection_alert_time>0</agents_disconnection_alert_time>
  <update_check>yes</update_check>
</global>

<alerts>
  <log_alert_level>3</log_alert_level>
  <email_alert_level>8</email_alert_level>
</alerts>
```

The brute-force alert was successfully received in the Gmail inbox with event details, confirming that the end-to-end detection and email alerting workflow was working correctly

Brute Force Attack Simulation:

```
Command Prompt
C:\Secure_assests>runas /user:ghost cmd
Enter the password for ghost:
Attempting to start cmd as user "DESKTOP-OVDU9P7\ghost" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Secure_assests>runas /user:ghost cmd
Enter the password for ghost:
Attempting to start cmd as user "DESKTOP-OVDU9P7\ghost" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Secure_assests>runas /user:ghost cmd
Enter the password for ghost:
Attempting to start cmd as user "DESKTOP-OVDU9P7\ghost" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Secure_assests>runas /user:ghost cmd
Enter the password for ghost:
Attempting to start cmd as user "DESKTOP-OVDU9P7\ghost" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Secure_assests>runas /user:ghost cmd
Enter the password for ghost:
Attempting to start cmd as user "DESKTOP-OVDU9P7\ghost" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Secure_assests>runas /user:ghost cmd
Enter the password for ghost:
Attempting to start cmd as user "DESKTOP-OVDU9P7\ghost" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Secure_assests>runas /user:ghost cmd
Enter the password for ghost:
Attempting to start cmd as user "DESKTOP-OVDU9P7\ghost" ...
RUNAS ERROR: Unable to run - cmd
1909: The referenced account is currently locked out and may not be logged on to.
```

Detection in Wazuh Dashboard:

W

Threat Hunting

qasim-pc

a

0

21:00 22:00 23:00 00:00 01:00 02:00 03:00 04:00 05:00 06:00 07:00

timestamp per 10 minutes

58 hits

Jan 13, 2026 @ 20:05:28.303 - Jan 14, 2026 @ 07:53:01.253

Export Formatted

Reset view

730 available fields

Columns

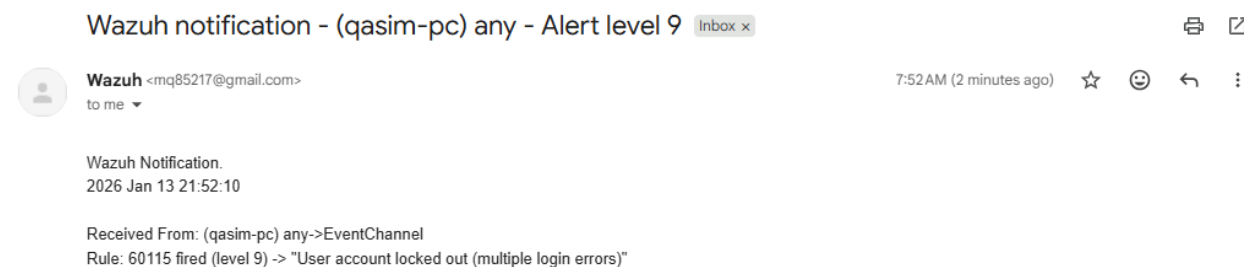
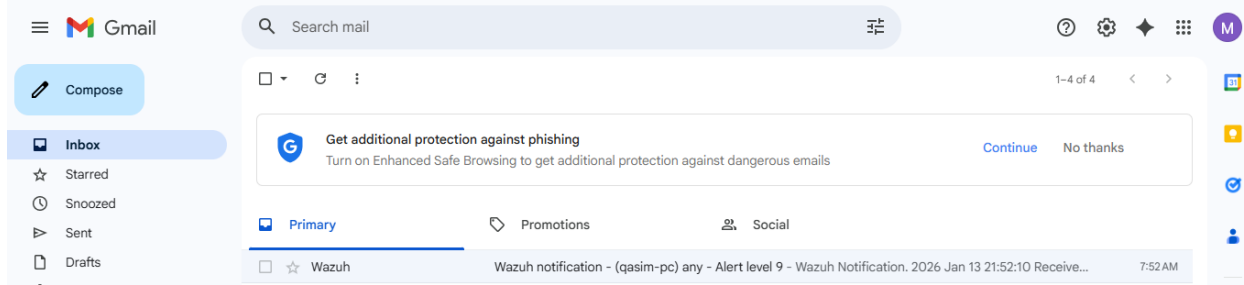
Density

1 fields sorted

Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Jan 14, 2026 @ 07:52:10.1...	qasim-pc	Auth: Failed logon detected (single event)	3	100199
Jan 14, 2026 @ 07:52:10.1...	qasim-pc	User account locked out (multiple login errors)	9	60115
Jan 14, 2026 @ 07:52:04.4...	qasim-pc	Auth: Failed logon detected (single event)	3	100199
Jan 14, 2026 @ 07:52:01.0...	qasim-pc	Auth: Failed logon detected (single event)	3	100199
Jan 14, 2026 @ 07:51:58.1...	qasim-pc	Auth: Failed logon detected (single event)	3	100199
Jan 14, 2026 @ 07:51:55.5...	qasim-pc	Auth: Multiple failed logons in 5 minutes (possible brute force/spray)	7	100200
Jan 14, 2026 @ 07:51:52.6...	qasim-pc	Auth: Failed logon detected (single event)	3	100199
Jan 14, 2026 @ 07:51:49.0...	qasim-pc	Auth: Failed logon detected (single event)	3	100199
Jan 14, 2026 @ 07:51:45.7...	qasim-pc	Auth: Failed logon detected (single event)	3	100199
Jan 14, 2026 @ 07:51:40.9...	qasim-pc	Auth: Failed logon detected (single event)	3	100199

Alert Email:



8.Conclusion

This project provided hands-on experience in setting up and working with a SIEM environment using wazuh. Through this lab, endpoint monitoring was successfully implemented, logs were collected and verified, and custom detection rules were created to identify security-relevant activity.

Different security scenarios were tested by performing actions on the endpoint and observing how they were detected and correlated into alerts. Email notifications were also configured to ensure important events were delivered outside the dashboard.

Overall, this project helped develop a practical understanding of how security events are monitored, detected, and reported in a real monitoring environment, moving beyond theory to hands-on implementation.